



Bishram Acharya

Computer Engineer
Pulchowk Campus, IOE, Lalitpur

+977-9860834906

bisram019@gmail.com

077bct023.bishram@pcampus.edu.np

[Github-\(bishram-acharya\)](https://github.com/bishram-acharya)

[LinkedIn-\(bishram-acharya\)](https://www.linkedin.com/in/bishram-acharya)

EDUCATION

•Pulchowk Campus, IOE, Lalitpur

2025 March

Bachelor's in Computer Engineering.

Score: 80%

- * Relevant Courses: Artificial Intelligence, Probability and Statistics, Data Structures and Algorithms, DevOps Fundamentals, Software Engineering, Theory of Computation, Discrete Structures, Microprocessor, C and C++ Programming, Computer Organization and Architecture, Mathematics

EXPERIENCES

•Research Intern at NAAMII

September 2024-present

Transforming Global Health with AI (TOGAI) Lab

- Built pipeline for ISIC skin lesion classification challenge
- Currently engaging with papers in the intersection of Vision Language Models and Active Learning

•Freelancer

Nov 2023-present

Upwork

- Built a Python Application for automatic news extraction and display from skynews.au. Used **Python, Html, CSS**
- Built a data scraper for a real estate website. Used **Python, Selenium**
- [Upwork Profile](#)

PROJECTS

•Cross Platform Fingerprint Matching Using CNN

Third Year Minor Project

- Trained a siamese network made of a couple of VGG-16s and DenseNet to determine whether the two given fingerprints are match or no match. This project aimed to observe the usefulness of VGG16 and DenseNet as a Siamese network for effective matching of contactless and contact-based fingerprints.
- Tools & technologies used: Python, TensorFlow, Numpy, Pillow

•AI Generated Image Detection

Final Year Major Project

- Implemented a frequency-aware deepfake detection model, FreqNet, which uses convolutional layers within the frequency space and incorporates High-Frequency Representation of Image (HFRI), High-Frequency Representation of Features (HFRF), and Frequency Convolutional Layers (FCL) to enhance detection accuracy and generalize across various deepfake methods by capturing subtle, high-frequency details.
- Tools Technologies Used: Python, PyTorch, Numpy, OpenCV

•Bitcoin Portfolio Analysis: Risk, Returns & Optimization

Analysis of 8-year performance of Bitcoin vs. S&P 500 and gold

- Analyzed Bitcoin's risk and return profile in comparison to traditional assets like the SP 500 and Gold.
- Evaluated the impact of adding Bitcoin to a traditional portfolio on overall performance and risk-adjusted returns.
- Tools & technologies used: Pandas, NumPy, Seaborn, Matplotlib, Sklearn

•Skin Cancer Disease Classification

ISIC 2019 Challenge

- Developed a deep learning model using ResNet-18 for the classification of skin lesions in dermoscopic images as part of the ISIC 2019 Challenge.
- Tools Technologies Used: Python, PyTorch, Tqdm, Pillow, Scikit-learn

•Logistic Regression from Scratch

Supervised Learning Project

- Implemented Logistic Regression from scratch and solve a binary classification problem on heart disease dataset.
- Tools & technologies used: Python, Pandas, NumPy, Seaborn, Sklearn, Sklearn metrics

•Energy Demand Prediction

Time Series Project

- Predict the future energy demand given the weather conditions as input.
- Tools & technologies used: Pandas, NumPy, Seaborn, Sklearn, RandomForestRegressor, Sklearn metrics

CERTIFICATIONS AND COURSES

- **Deep Learning Specialization (Certificate)**

Coursera-Andrew Ng

115 Hours

- [Neural Networks and Deep Learning](#): Vectorization, backpropagation, deep neural networks, [Computer Vision](#); [Hyperparameter Tuning](#), [Regularization and Optimization](#), L2 Regularization, Dropout; CNN layers, transfer learning, multi-class classification, object detection, [Sequence Models](#): RNNs, LSTMs, GRUs for NLP tasks; attention mechanism for speech recognition, optimizing ML workflows, [Structuring ML Projects](#), end-to-end deep learning.

- **Complete Python Developer Course (Certificate)**

Udemy

31 Hours

- Data Structures, Object-Oriented Programming with Python, Functional Programming with Python, Lambdas, Decorators, Generators, Testing in Python, Debugging, Error Handling, Regular Expressions, Comprehensions, Modules, Virtual Environments, Developer Environments

- **Artificial Intelligence Course - Samsung Innovation Campus (Certificate)**

Samsung

6 months

- Math and statistics, data preprocessing, visualization, machine learning (classification, clustering, anomaly detection), NLP, deep learning (CNN, RNN, GANs), dimensionality reduction, neural networks, and language models.

- **Complete Data Science & Machine Learning Bootcamp (Certificate)**

Udemy

44 Hours

- Data Exploration and Visualizations, Neural Networks and Deep Learning, Model Evaluation and Analysis, Python, Tensorflow 2.0, Numpy, Pandas, Scikit-Learn, Data Science and Machine Learning Projects and Workflows, Data Visualization in Python with Matplotlib and Seaborn, Transfer Learning

- **Generative AI with Large Language Models (Certificate)**

AWS/DeepLearning.AI

16 Hours

- LLM architecture and design, transformer models and attention mechanisms, domain-specific adaptation of models, efficient multi-GPU compute strategies, parameter-efficient fine-tuning (PEFT), LoRA and soft prompt techniques, reinforcement learning from human feedback (RLHF), model evaluation metrics and benchmarks, LLM applications.

- **Custom Models, Layers, and Loss Functions with TensorFlow (Certificate)**

Coursera-DeepLearning.AI

31 Hours

- Functional API and Sequential API, Flexibility of Functional API, Building a Siamese Network, Custom Loss Functions, Building Custom Loss Functions, Contrastive Loss Function, Custom Layers, Building Custom Layers, Extending TensorFlow Model Class, Building a ResNet Model, Custom Callbacks, Implementing Custom Callbacks to Detect Overfitting